

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

BELAGAVI - 590 018



**PROJECT WORK
REPORT**

On

Submitted in the partial fulfillment of the requirements for the award of the degree of

**BACHELOR OF ENGINEERING
in
ELECTRONICS & COMMUNICATION
SMART VEHICLE IGNITION SYSTEM**

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Certificate

Certified that the Project work entitled “*Smart Vehicle Ignition System*” carried out by Ms.Bhagya D B, USN 4BD21EC016, Ms.Rani C O, USN 4BD21EC078, Ms.Sindhu B S, USN 4BD21EC100, Ms.Srushti S G, USN 4BD20EC106, bonafide students of this institution in partial fulfillment for the award of degree of **Bachelor of Engineering in Electronics & Communication** by **Visvesvaraya Technological University**, Belgaum during the academic year **2024-25**. It is certified that all corrections / suggestions indicated for internal evaluation have been incorporated in the report deposited in the departmental library. The project report has been approved as it satisfies the academic requirements in respect of Project work prescribed for the said degree.

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MISSION OF THE INSTITUTE

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PROJECT WORK: (21ECP76)

Course Learning Objectives:

This course will enable us to:

- Prepare with foundation knowledge in a project domain through designing, and implementing.
- Introduce to various emerging fields in electronics and communication.
- Provide an opportunity to exercise the creative and innovative qualities, to become technopreneurs in group project environment.
- Have hands-on experience in the related field so that we can integrate and reinforce what has been taught in the classroom.

ABSTRACT

The project presents a multi-layered Smart Vehicle Ignition System that enhances vehicle security and safety by integrating RFID authentication, facial recognition, and fingerprint sensing. Only upon successful verification through all three biometric and identification methods does the vehicle ignition system activate, providing a robust defense against unauthorized access.

Once the vehicle is started, alcohol detection and tilt sensing mechanisms are initiated to ensure the driver's sobriety and monitor the vehicle's orientation for potential accidents or rollovers. The alcohol sensor checks the driver's breath for traces of alcohol, and if detected beyond a safe threshold, the system triggers an alert and may shut down the motor to prevent impaired driving. The tilt sensor monitors unusual tilts or rollovers and can alert emergency services or shut down the system as a preventive measure.

This comprehensive system aims to enhance vehicular security, ensure driver responsibility, and provide real-time safety monitoring, making it highly suitable for personal vehicles, public transport, and fleet management.

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CHAPTER 1

INTRODUCTION

Vehicle theft continues to be a pervasive issue globally, leading to significant financial losses and emotional distress for owners. Conventional vehicle security systems, often relying on singular authentication methods like physical keys or basic alarms, are increasingly susceptible to circumvention by determined thieves. Furthermore, critical driver safety aspects, such as preventing impaired driving due to alcohol consumption and providing timely alerts in case of accidents, are often not integrated into standard security packages. This creates a demand for more holistic, robust, and intelligent solutions that not only prevent unauthorized access but also enhance overall vehicle and driver safety. The modern vehicle owner also seeks greater connectivity and control, desiring real-time updates and the ability to interact with their vehicle remotely.

1.2 Significance of the Project:

This project, "Multi-Level Security System for Vehicles," directly addresses the identified needs by proposing and implementing a comprehensive, layered security and safety architecture. Its significance lies in:

- **Enhanced Security:** By integrating three distinct authentication tiers – face recognition, fingerprint scanning, and RFID-based license validation – the system establishes a formidable barrier against unauthorized vehicle operation. The multi-modal approach ensures that compromising a single layer does not grant access.
- **Proactive Safety:** The inclusion of an alcohol detection sensor to prevent engine ignition by an intoxicated driver and an accident detection sensor (tilt sensor) to potentially alert authorities or owners significantly enhances driver and passenger safety.
- **IoT Integration & Remote Control:** Leveraging the ESP32 microcontroller and the Blynk IoT platform, the system provides vehicle owners with real-time notifications (e.g., engine activation) on their smartphones and empowers them with remote control capabilities, such as disabling the engine if unauthorized use is suspected

- **IoT Integration & Remote Control:** Leveraging the ESP32 microcontroller and the Blynk IoT platform, the system provides vehicle owners with real-time notifications (e.g., engine activation) on their smartphones and empowers them with remote control capabilities, such as disabling the engine if unauthorized use is suspected.
- **User-Friendly Interface:** An LCD display provides clear, real-time status updates and instructions to the user during the authentication process. This project aims to deliver an advanced, intelligent, and interactive security and safety solution, setting a higher standard for vehicle protection and driver well-being.

CHAPTER 2

LITERATURE SURVEY

1. Face Recognition Based Car Ignition System with Anti-Theft Security using GSM Technology.

Ahmed, S., & Hasan, M. K.

This paper discusses a system using face recognition for car ignition and GSM for sending alerts. It highlights the feasibility of face recognition as an initial security layer. *Our project expands on this by adding further biometric and RFID layers, and using Blynk for more interactive IoT capabilities instead of just GSM alerts.*

2. A Multi-Layered Vehicle Security System using Fingerprint and RFID Technology.

Kumar, P., & Singh, R.

This work details a system combining fingerprint biometrics and RFID for vehicle access, emphasizing the increased security from multiple authentication factors. *Our project incorporates these layers and adds face recognition as an initial, non-contact layer, plus safety sensors.*

3. IoT Based Smart Vehicle Alcohol Detection and Accident Alert System.

Patel, R., Shah, M., & Makwana, R.

This research focuses on an IoT system for vehicles that detects alcohol and alerts authorities in case of accidents using sensors and GPS/GSM. *Our project integrates similar alcohol and accident detection functionalities but uses Blynk for notifications and owner control, and embeds these within a broader access control system.*

4. Robust Real-Time Face Detection.

Viola, P., & Jones, M. J.

This foundational paper describes the Haar-like features and AdaBoost-based cascade classifier for rapid and robust face detection, which has become a standard in the field. *Our project utilizes this Haar Cascade algorithm for the initial face detection stage within the Python-based face recognition module.*

5. Face Description with Local Binary Patterns: Application to Face Recognition.

Ahonen, T., Hadid, A., & Pietikäinen, M.

This paper introduces Local Binary Patterns (LBP) and its histogram-based variant (LBPH) as powerful and computationally efficient descriptors for face recognition, known for their robustness to illumination changes. *Our project employs the LBPH algorithm for the core face recognition process after a face is detected.*

6. ESP32 Based Smart Security System with Blynk App Integration.

Manivannan, S., et al.

This paper showcases the use of ESP32 for security applications, interfacing various sensors and utilizing the Blynk platform for remote monitoring and control, demonstrating the ESP32's suitability for IoT-based security projects. *Our project leverages the ESP32 and Blynk in a similar manner for its IoT functionalities and sensor integration.*

Summary of literature survey

Ref. Paper authors / year	Objectives / Issues addressed	Parameter Evaluated	Adaptive Technique	Drawbacks	Open issue/ Improvements
Ahmed & Hasan, 2017	Car ignition security using face recognition and GSM alerts	Face recognition accuracy, GSM alerts	Face recognition + GSM module	Limited to single biometric and SMS alerts only	Integrate multi-factor authentication and IoT-based notifications
Kumar & Singh, 2019	Enhance vehicle access security using fingerprint + RFID	Authentication accuracy, Security strength	Fingerprint + RFID-based access	No remote monitoring, lacks additional safety checks	Add face recognition as initial layer and IoT-based control
Patel, Shah & Makwana, 2020	Alcohol detection and accident alerts using IoT	Detection accuracy, Alert response	Alcohol sensor + GSM/GPS alert system	Limited to safety detection only, lacks access control	Integrate Blynk app for interactive alerts and owner control
Viola & Jones, 2004	Robust real-time face detection	Detection speed, Accuracy	Haar Cascade (AdaBoost with Haar-like features)	Sensitive to pose variations and occlusions	Combine with LBPH for recognition stage
Ahonen, Hadid & Pietikäinen, 2006	Efficient face recognition under illumination variation	Recognition accuracy, Speed	Local Binary Pattern Histogram (LBPH)	Limited performance in complex environments	Improve with hybrid recognition methods
Manivannan et al., 2021	IoT security with ESP32 and Blynk integration	IoT connectivity, Remote monitoring	ESP32 microcontroller + Blynk IoT platform	Dependent on stable internet connection	Add offline control and expand sensor integration

CHAPTER 3

PROBLEM IDENTIFICATION

Current vehicle security and safety systems often exhibit several limitations:

1. **Single-Factor Vulnerability:** Many systems rely on a single point of authentication (e.g., key, basic key fob), which, if compromised (stolen, duplicated, hacked), grants full access to the vehicle.
2. **Ease of Bypass:** Traditional car alarms can often be disabled quickly, and methods to bypass simple electronic immobilizers are known.
3. **Reactive Rather Than Proactive Safety:** Most standard systems lack features to proactively prevent unsafe driving conditions (like drunk driving) or to automatically alert in case of an accident.
4. **Limited Owner Awareness and Control:** Owners often have minimal real-time information about their vehicle's status when they are away from it and lack immediate remote control options in case of theft or unauthorized use beyond a basic alarm sound.
5. **No Intrinsic Driver Validation:** Standard systems do not verify if the person attempting to operate the vehicle is indeed authorized or possesses a valid license for that specific context (though RFID here is a simplified proxy).

This project aims to directly address these identified problems by developing a multi-faceted system that is significantly harder to compromise, actively contributes to driver safety, and provides enhanced owner interaction and control.

CHAPTER 4

OBJECTIVES

The primary objectives of this project are:

1. To design and implement a robust multi-level authentication system for vehicle ignition, comprising:
 - **Level 1:** Face recognition using Python (OpenCV, Haar Cascades for detection, LBPH for recognition) executed on a laptop, with results communicated to an ESP32.
 - **Level 2:** Fingerprint verification using an optical fingerprint sensor interfaced with the ESP32.
 - **Level 3:** RFID tag scanning for user/license validation (as a proxy for license check) using an RFID reader interfaced with the ESP32.
2. To integrate essential driver safety features:
 - a. An alcohol detection system using an MQ-3 sensor to prevent engine start if the driver's breath alcohol content exceeds a predefined threshold.
 - b. An accident detection system using a tilt sensor to identify potential collisions or rollovers.
3. To develop an IoT-based remote monitoring and control system using the ESP32's Wi-Fi capabilities and the Blynk platform, enabling:
 - a. Real-time notifications to the vehicle owner's smartphone upon successful engine activation.
 - b. The ability for the owner to remotely disable the vehicle's engine via the Blynk application.
4. To provide clear, step-by-step instructions and status feedback to the user via an LCD display.
5. To ensure that the vehicle's engine (simulated by a DC motor) can only be activated after all security layers and safety checks are successfully passed.
6. To establish reliable serial communication between the laptop (Python face recognition module) and the ESP32 (main control unit).

CHAPTER 5

METHODOLOGY

The project employs a layered, modular approach to achieve its objectives. The methodology involves distinct hardware and software components working in concert.

System Architecture Overview:

The system is architecturally divided into two main processing units:

- 1. Laptop (Face Recognition Unit):** Handles the computationally intensive face recognition task using its built-in webcam and Python with OpenCV.
- 2. ESP32 Microcontroller (Main Control & IoT Unit):** Manages all subsequent authentication layers (fingerprint, RFID), sensor inputs (alcohol, tilt), user interface (LCD), engine control (relay), and IoT communication (Blynk).

Communication between the laptop and ESP32 occurs via serial (USB) connection.

5.1 BLOCK DIAGRAM

Fig 5.1.1 shows the block diagram of the proposed system, where the ESP32 microcontroller interfaces with sensors and modules for vehicle security and safety. It manages authentication, accident/alcohol detection, and IoT-based notifications.

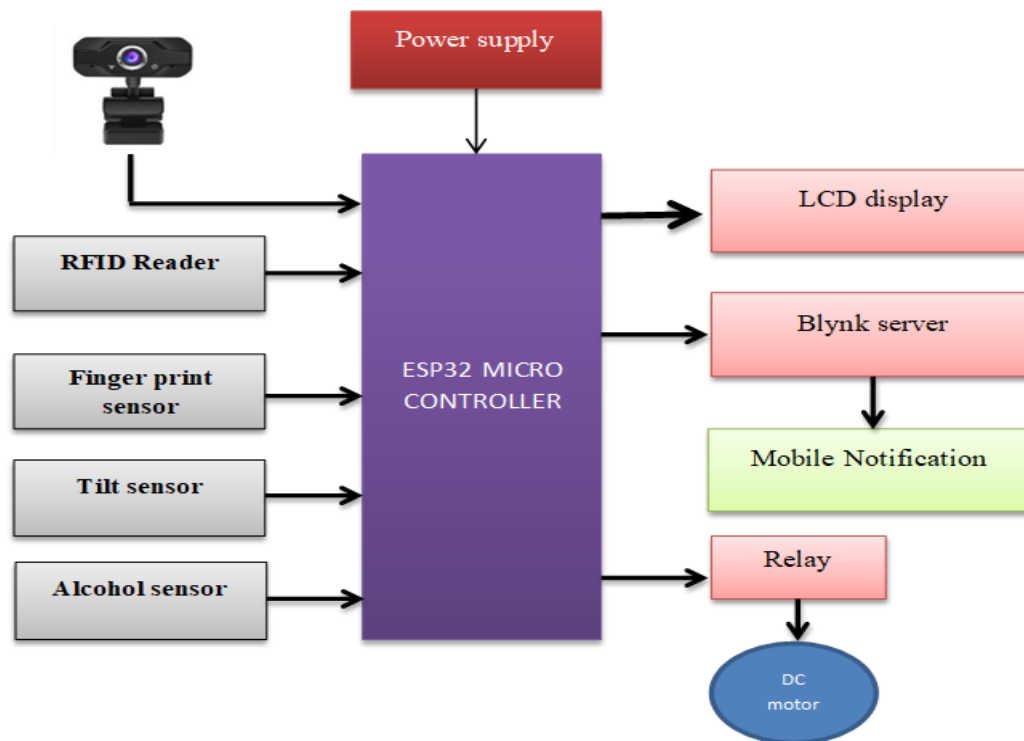


Figure 5.1.1: Block diagram

Laptop Computer: The laptop's webcam captures the user's face. Python scripts using OpenCV libraries (Haar Cascades for detection, LBPH for recognition) process the image. If a registered face is recognized, a success signal (e.g., "FACE_OK") is sent via serial (USB) to the ESP32.

ESP32 Microcontroller: This is the core controller.

- It receives the face recognition status from the laptop.
- Prompts the user for fingerprint scanning via the LCD and verifies the fingerprint using the connected sensor.

Prompts for RFID tag scanning and validates the tag.

- Reads data from the alcohol sensor to check for intoxication.
- Monitors the tilt sensor for accident detection.
- Displays all prompts, status messages, and alerts on the LCD.
- If all checks pass, it activates a relay to turn ON the DC motor (simulating the vehicle engine).
- Uses its built-in Wi-Fi to connect to the Blynk server, send notifications (e.g., "Engine ON") to the owner's smartphone app, and receive remote commands (e.g., "Engine OFF").

Blynk Server & App: Acts as the intermediary for IoT communication. The ESP32 sends data to and receives commands from the owner's Blynk app on their smartphone through the Blynk cloud server.

5.2 FLOW CHART

The fig 5.2.1 shows the flowchart of the proposed system, which illustrates the step-by-step functioning of driver authentication and safety monitoring. It integrates face recognition, RFID, fingerprint, alcohol detection, and tilt sensors for decision-making and control.

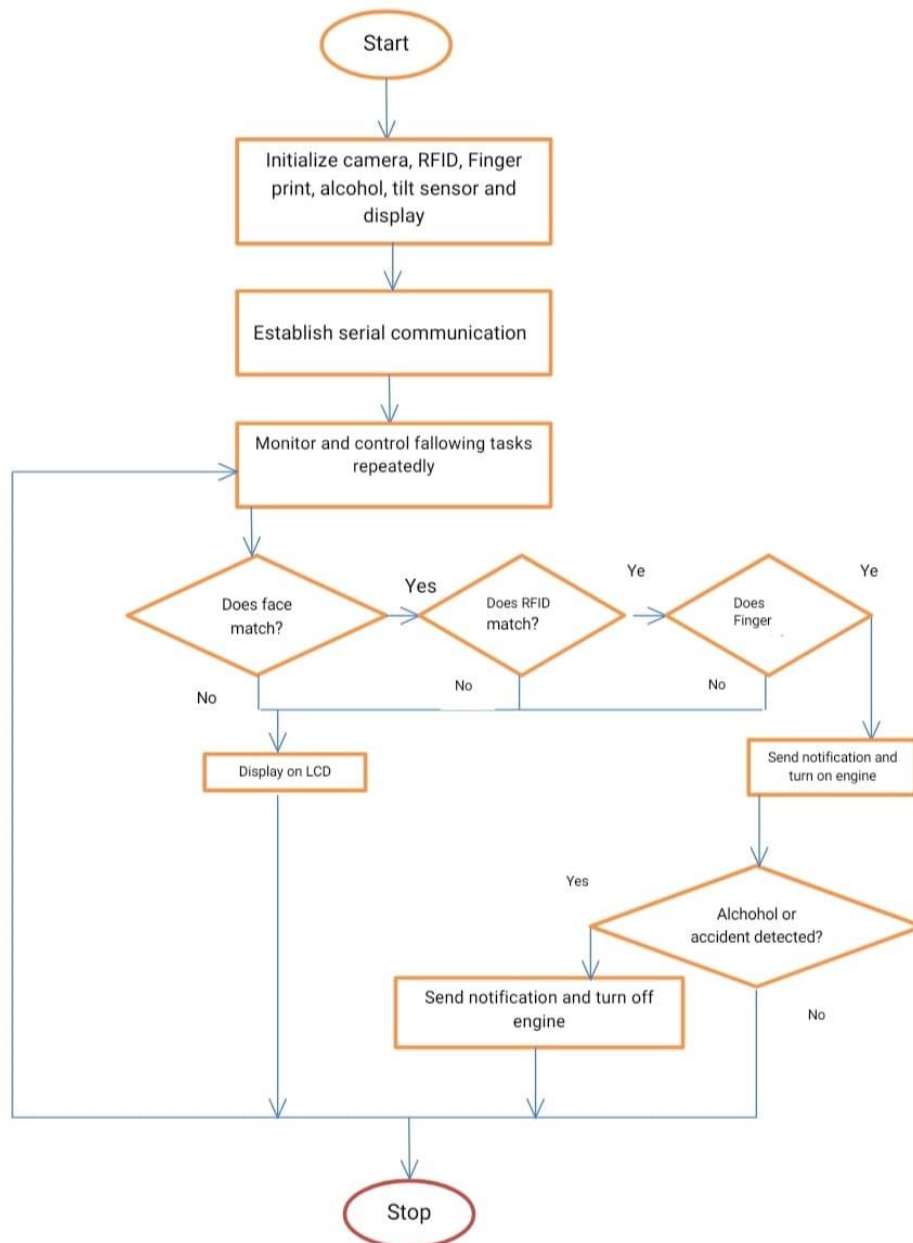


Fig 5.2.1 : Flow diagram

The system begins by initializing all necessary hardware components including the camera, sensors, and display. It then enters a continuous loop where it first checks if the face matches a stored profile. If successful, it proceeds to verify the RFID tag and then the fingerprint. If all three authentication steps pass, the system sends a notification and starts the engine. After ignition, it monitors for alcohol presence or accident detection via tilt sensors. If any such event is detected, the system immediately sends a notification and turns off the engine to ensure safety. This structured approach enhances access vehicle.

CHAPTER 6

HARDWARE REQUIREMENTS

6.1 ESP32 BOARD



Figure 6.1.1: ESP32 Board

ESP32 Development Board (e.g., ESP32-WROOM-32 based):

- **Specification:** Dual-core Tensilica LX6 microprocessor, integrated Wi-Fi (802.11 b/g/n) and Bluetooth (v4.2 BR/EDR & BLE), multiple GPIOs, 12-bit ADCs, SPI, I2C, UART interfaces.
- **Role:** Main control unit for authentication (post-face), sensors, LCD, relay, and IoT communication.

6.2 Laptop with Built-in Webcam:

- **Specification:** Standard personal computer with an integrated camera (e.g., 720p or 1080p resolution). Sufficient processing power (e.g., Intel Core i3/i5 or equivalent, 4GB+ RAM) to run Python and OpenCV.
- **Role:** Captures user's face, performs face detection and recognition.

6.3 Fingerprint Sensor (e.g., R307/R503 Optical Module):

- **Specification:** Optical sensing technology, UART interface, typical storage capacity for multiple fingerprint templates.

- Role: Second-level biometric authentication.



Figure 6.3.1: Fingerprint Sensor

6.4 RFID Reader Module (MFRC522):

- **Specification:** Operates at 13.56 MHz, SPI interface, reads ISO/IEC 14443-A type tags/cards.
- **Role:** Third-level authentication (user/license validation proxy).



Figure 6.4.1: RFID Reader

6.5 Alcohol Sensor (MQ-3):



Figure 6.5.1: Alcohol sensor(MQ3)

- **Specification:** Semiconductor sensor sensitive to alcohol vapor, provides an analog voltage output proportional to concentration. Requires a warm-up time.
- **Role:** Detects alcohol on the driver's breath for safety.

6.6 Tilt Sensor (e.g., SW-520D or MPU6050 Gyro/Accelerometer):



Figure 6.6.1: Tilt Sensor

- **Specification (SW-520D):** Simple roller-ball mechanism, digital output (ON/OFF state change upon tilting beyond a certain angle).
- **Specification (MPU6050):** 3-axis accelerometer and 3-axis gyroscope, I2C interface. More sophisticated, can detect angles and impacts. (If using MPU6050, complexity increases). *Assuming SW-520D for simplicity based on "tilt sensor."*
- **Role:** Detects significant tilting of the vehicle, indicating a potential accident.

6.7 LCD Display (16x2 I2C):



Figure 6.7.1: LCD Display

- **Specification:** 16 characters per line, 2 lines. I2C interface (typically PCF8574 based backpack) for reduced pin usage.
- **Role:** User interface for prompts, status messages, and alerts.

6.8 Relay Module (5V Single Channel):



Figure 6.8.1: Relay Module

- **Specification:** Electromechanical switch, typically rated for 5V DC coil voltage, capable of switching AC or higher DC loads (e.g., 10A/250VAC or 10A/30VDC).
- **Role:** Switches the DC motor (simulating vehicle engine) ON/OFF based on ESP32 signal.

6.9 DC Motor (e.g., 3V-6V or 6V-12V Gear Motor):



Figure 6.9.1: DC motor

- **Specification:** Standard DC motor, voltage and current ratings compatible with relay and power supply.
- **Role:** Simulates the vehicle's engine operation.

6.10 Connecting Wires (Jumper Wires), Breadboard.

6.11 Power Supply:

Appropriate power supply for ESP32, sensors, and DC motor (may require separate supplies or a robust single supply with voltage regulation).

6.12 Complete schematic diagram and its description.

Figure 6.12.1 shows the schematic diagram depicts a vehicle access system using an ESP32 MCU, with inputs from an Alcohol sensor, Tilt sensor, RFID scanner, and Fingerprint sensor. It controls a relay to activate the motor based on sensor readings, and displays system status on the LCD.

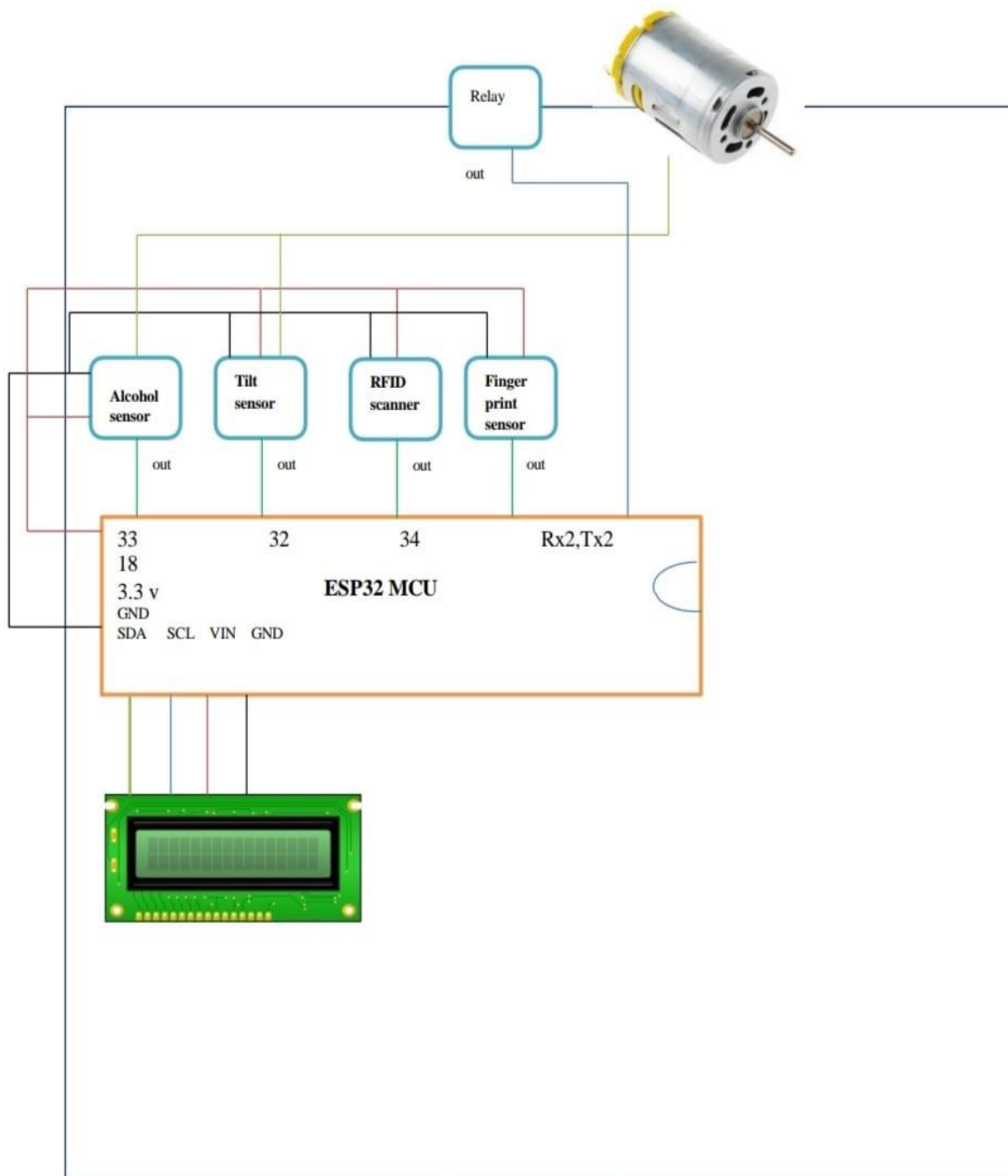


Fig 6.12.1: Schematic Diagram

- The USB connection to the laptop would be indicated.
- The Fingerprint sensor's TX, RX, VCC, GND pins would be connected to the designated UART2 pins and power rails of the ESP32.
- The MFRC522 RFID reader's SPI pins (MOSI, MISO, SCK, CS, RST) and power pins would be connected to the corresponding HSPI/VSPI pins and 3.3V/GND rails on the ESP32.
- The MQ-3 Alcohol Sensor's AOUT pin would connect to an ADC pin on the ESP32, with VCC and GND to 5V and GND.
- The Tilt Sensor's output pin would connect to a digital input pin on the ESP32, with VCC and GND to 3.3V/5V and GND.
- The I2C LCD's SDA and SCL pins would connect to the ESP32's I2C pins (GPIO21, GPIO22), and VCC/GND to 5V/GND.
- The Relay Module's IN pin would connect to a digital output pin on the ESP32, with its VCC/GND connected to 5V/GND.
- The DC Motor would be shown connected across the COM and NO terminals of the relay, with its own power source (e.g., a battery pack or bench power supply) completing the circuit through these relay contacts.
- Power distribution, including any necessary voltage regulators (if ESP32 5V output is insufficient for all 5V peripherals), pull-up/pull-down resistors (if not internally handled by modules), and decoupling capacitors, would also be depicted.

CHAPTER 7

SOFTWARE REQUIREMENTS

7.1 Face Detection Algorithm (Haar Cascades - Python, OpenCV):

- **Description:** The Haar Cascade classifier is an effective object detection method. It's trained on a large set of positive images (faces) and negative images (non-faces). It uses Haar-like features, which are differences between sums of pixel intensities in rectangular regions. These features are processed by a cascade of classifiers. Each stage in the cascade is a collection of weak classifiers. If a region of an image passes all stages, it's classified as a face.
- **Example:** When the Python script receives a frame from the webcam, it converts it to grayscale. The detect MultiScale function is then applied, which scans the image at multiple scales. It returns a list of rectangles (x, y, width, height) corresponding to detected faces.

7.2 Face Recognition Algorithm (LBPH - Local Binary Patterns Histograms - Python, OpenCV):

- **Description:**
 1. **Training:**
 - For each training image of a known person:
 - The face is detected (using Haar Cascades).
 - The detected face region is converted to grayscale.
 - The LBP operator is applied: For each pixel, a binary code is generated by comparing its intensity with its 8 neighbours. If a neighbour is brighter or equal, it's 1; otherwise, it's 0. This 8-bit binary number is converted to decimal.
 - The face image is divided into a grid of small cells (e.g., 8x8 cells).
 - For each cell, a histogram of these LBP decimal values is computed.
 - These histograms are concatenated to form a single, long feature vector (the LBPH) for that face.
 - This feature vector is stored along with a label (person's ID or name).

7.3 Recognition:

When a new face is presented:

- It's detected and the LBPH feature vector is computed as above.
- This new vector is compared to all stored feature vectors from the training set (e.g., using Chi-square distance or absolute difference).
- The label of the training image with the closest feature vector is assigned to the new face. A confidence score (related to the distance) is also calculated. If the confidence is below a certain threshold (meaning the distance is small enough), the recognition is considered successful.
- **Example:** If a pixel has neighbors resulting in binary 10110010, this translates to decimal 178. This value contributes to the histogram of its local cell. The combined histograms robustly represent facial texture.

7.4 Serial Communication Protocol (Python PySerial & ESP32 Arduino Serial):

- **Description:** A simple character-based or string-based protocol is used for communication between the Python script on the laptop and the ESP32. The laptop acts as the sender after face recognition, and the ESP32 as the receiver.
- **Example:**
 - Python (Sender): `ser.write(b'FACE_OK\n')` if face recognized, or `ser.write(b'FACE_FAIL\n')` if not.
 - ESP32 (Receiver): Reads incoming serial data until a newline character `\n`. Compares the received string to "FACE_OK" to proceed.

7.5 Fingerprint Matching Algorithm (Sensor Internal):

- **Description:** The fingerprint sensor (e.g., R307) has its own internal microprocessor and algorithms for image capture, minutiae extraction (finding unique points like ridge

endings and bifurcations), template generation, and matching. The ESP32 sends commands to the sensor (e.g., "capture image," "generate template," "search database") via UART.

- **Example:** ESP32 sends a command to enroll a finger. User places finger. Sensor captures image, extracts minutiae, creates a template, and stores it with an ID. For verification, user places finger, sensor captures, creates a template, and compares it against stored templates. It returns a match/no-match result to ESP32.

7.6 RFID Tag Validation (ESP32):

- **Description:** The MFRC522 RFID reader detects nearby 13.56 MHz tags. When a tag is in range, the ESP32 commands the reader (via SPI) to read the tag's Unique ID (UID). The ESP32 then compares this UID against a predefined list of authorized UIDs.
- **Example:** `authorized_rfid_uids = ["0xAB 0xCD 0xEF 0x12", "0x34 0x56 0x78 0x90"]`. If scanned UID matches an entry, it's valid.

7.7 Alcohol Level Check (ESP32):

- **Description:** The MQ-3 sensor outputs an analog voltage that varies with alcohol concentration. The ESP32 reads this voltage using an Analog-to-Digital Converter (ADC) pin. This raw ADC value is then compared against a pre-calibrated threshold.
- **Example:** If `analogRead(MQ3_PIN) > ALCOHOL_THRESHOLD`, then alcohol is detected. (Calibration is crucial for accuracy).

7.8 Blynk IoT Communication Logic (ESP32):

- **Description:** The ESP32 uses the Blynk library to connect to the Blynk server via Wi-Fi. `Blynk.virtualWrite(V_PIN, value)`: Sends data from ESP32 to a widget (e.g., display, gauge) on the Blynk app associated with the specified virtual pin. `BLINK_WRITE(V_PIN)`: A function that is automatically called on the ESP32 when the state of a control widget (e.g., button, slider)

on the Blynk app associated with that virtual pin is changed by the user.

Blynk.notify("Message"): Sends a push notification to the owner's phone.

- **Example:**

To send engine status: Blynk.virtualWrite(V0, "Engine ON");

To receive remote engine OFF command:

```
BLYNK_WRITE(V1) { // V1 is a button widget in app
```

```
  int value = param.asInt();
```

```
  if (value == 1 && engine_is_active) { // Button pressed
```

```
    digitalWrite(RELAY_PIN, LOW); // Turn off engine
```

```
    Blynk.notify("Engine remotely disabled by owner!");
```

```
    // update LCD, internal state
```

```
  }
```

```
}
```

CHAPTER 8

RESULTS & DISCUSSION

8.1 Results of Each modules

- **Face Recognition Module (Laptop):**

1. **Expected Result:** The system should correctly recognize trained individuals with an accuracy of >90% under varying lighting conditions (though LBPH is somewhat robust to this) and minor variations in pose/expression. Recognition time per frame should be within 200-500ms on a moderately powered laptop.
2. **Testing:** Test with multiple images of registered users, images of unregistered users, and registered users under different conditions (glasses, hat, different lighting).
3. **Discussion:** False Positive Rate (FPR) and False Negative Rate (FNR) would be key metrics. Factors affecting performance (lighting, distance from camera, training dataset quality) would be discussed.

- **Fingerprint Authentication Module (ESP32):**

1. **Expected Result:** Successful enrollment of fingerprints and >95% accuracy in verifying registered users. Verification time should be < 2 seconds.
2. **Testing:** Enroll multiple fingers from different users. Test verification with correct fingers, incorrect registered fingers, and unregistered fingers. Test with varying finger pressure and condition (dry, slightly moist).
3. **Discussion:** Discuss reliability, ease of use, and any issues encountered with sensor sensitivity.

- **RFID Validation Module (ESP32):**

1. **Expected Result:** 100% accuracy in reading and validating registered RFID tags within the specified read range (typically a few cm for MFRC522). Read time should be almost instantaneous.
2. **Testing:** Test with registered tags, unregistered tags, and at varying distances/orientations.
3. **Discussion:** Confirm reliability and discuss practical read range.

- **Alcohol Detection Module (ESP32):**

1. **Expected Result:** The MQ-3 sensor should show a noticeable increase in analog output when exposed to alcohol vapor (e.g., from a test solution or breath after consuming alcohol, if ethically permissible and safe for testing). A clear threshold should be determinable to distinguish between sober and potentially intoxicated states.

2. **Testing:** Calibrate sensor with known non-alcohol air, then expose to controlled alcohol vapor. Observe ADC readings.

3. **Discussion:** Sensor warm-up time, sensitivity, stability, and the challenge of setting an accurate and reliable threshold for "intoxication" without professional calibration equipment would be discussed. Cross-sensitivity to other fumes could also be noted.

- **Accident Detection (Tilt Sensor) Module (ESP32):**

1. **Expected Result:** The tilt sensor should reliably change its digital state when the system is tilted beyond a predefined angle (e.g., >45 degrees).

2. **Testing:** Manually tilt the breadboard/prototype assembly to simulate an accident.

3. **Discussion:** Simplicity of the SW-520D vs. the potential for false positives (e.g., steep incline parking). An MPU6050 would offer more sophisticated detection but adds complexity.

- **Blynk IoT Integration (ESP32 & App):**

1. **Expected Result:** Real-time notifications for engine status ("Engine ON," "Accident," "Alcohol Detected") should be received on the Blynk app within a few seconds of the event. Remote engine OFF command from the app should promptly disable the DC motor.

2. **Testing:** Perform full system run-throughs, trigger events, and observe app notifications. Use the app to send commands.

3. **Discussion:** Reliability of Wi-Fi connection, latency of notifications/commands.

- **Overall System Integration and Performance:**

1. **Expected Result:** Smooth transition between authentication layers. Total time from initiating face recognition to engine start (for a successful run) should be acceptable (e.g., 10-20 seconds).

2. **Testing:** Multiple end-to-end test runs with various success/failure scenarios at each stage.

Test cases

Test Case ID	Test Scenario	Input	Expected Output	Pass/Fail Criteria
TC001	Face Recognition - Authorized User	Valid registered face	Authentication successful; proceed to fingerprint scan	System moves to next step (fingerprint)
TC002	Face Recognition - Unauthorized User	Unrecognized face	Authentication failed; access denied	Access is blocked
TC003	Fingerprint Scan - Valid	Valid registered fingerprint	Authentication successful; proceed to RFID check	System moves to next step (RFID)
TC004	Fingerprint Scan - Invalid	Invalid or unregistered fingerprint	Authentication failed; access denied	Access is blocked
TC005	RFID Check - Valid License Card	Valid RFID tag	Authentication successful; system ready to start engine	Access is granted
TC006	RFID Check - Invalid Tag	Unauthorized or unrecognized RFID tag	Authentication failed; access denied	Access is blocked
TC007	Alcohol Detection - No Alcohol	Breath sample = 0% BAC	Engine start allowed	System proceeds to ignition
TC008	Alcohol Detection - Alcohol Detected	Breath sample > threshold (e.g., 0.03% BAC)	Engine start blocked; alert triggered	Ignition prevented
TC009	Accident Detection	Tilt sensor detects sudden change (e.g., > threshold angle)	Accident alert sent via Blynk; system notifies owner	Notification received on Blynk app
TC010	Engine ON Notification	Successful 3-layer authentication	Real-time notification sent to owner via Blynk	Notification appears on smartphone
TC011	Remote Engine Disable via Blynk	Owner presses "Engine OFF" button on Blynk	Engine operation disabled	Engine stops remotely
TC012	LCD Display Feedback	Each security stage triggered	LCD shows real-time messages like "Scan Face", "Scan Finger", etc.	Appropriate message displayed at each stage
TC013	Power Interruption Recovery	System is restarted after power failure	System resumes and re-initiates authentication sequence	System functions as expected after reboot
TC014	Multiple Failed Attempts	Three consecutive failed login attempts	Lockout mechanism triggered or alert sent	System prevents further access temporarily
TC015	All Systems Go (Happy Path)	Valid face, fingerprint, RFID, no alcohol	Vehicle starts normally; all stages passed	Engine starts, user authenticated

8.2 Result

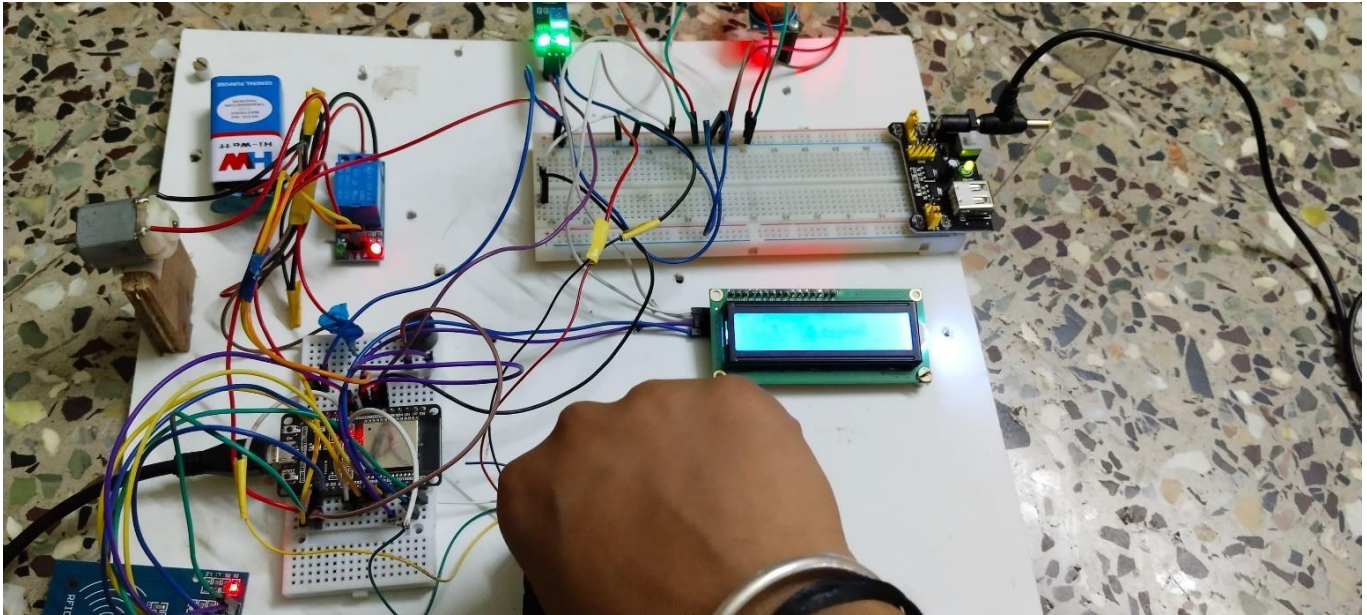


Figure 8.2.1: Waiting for Fingerprint

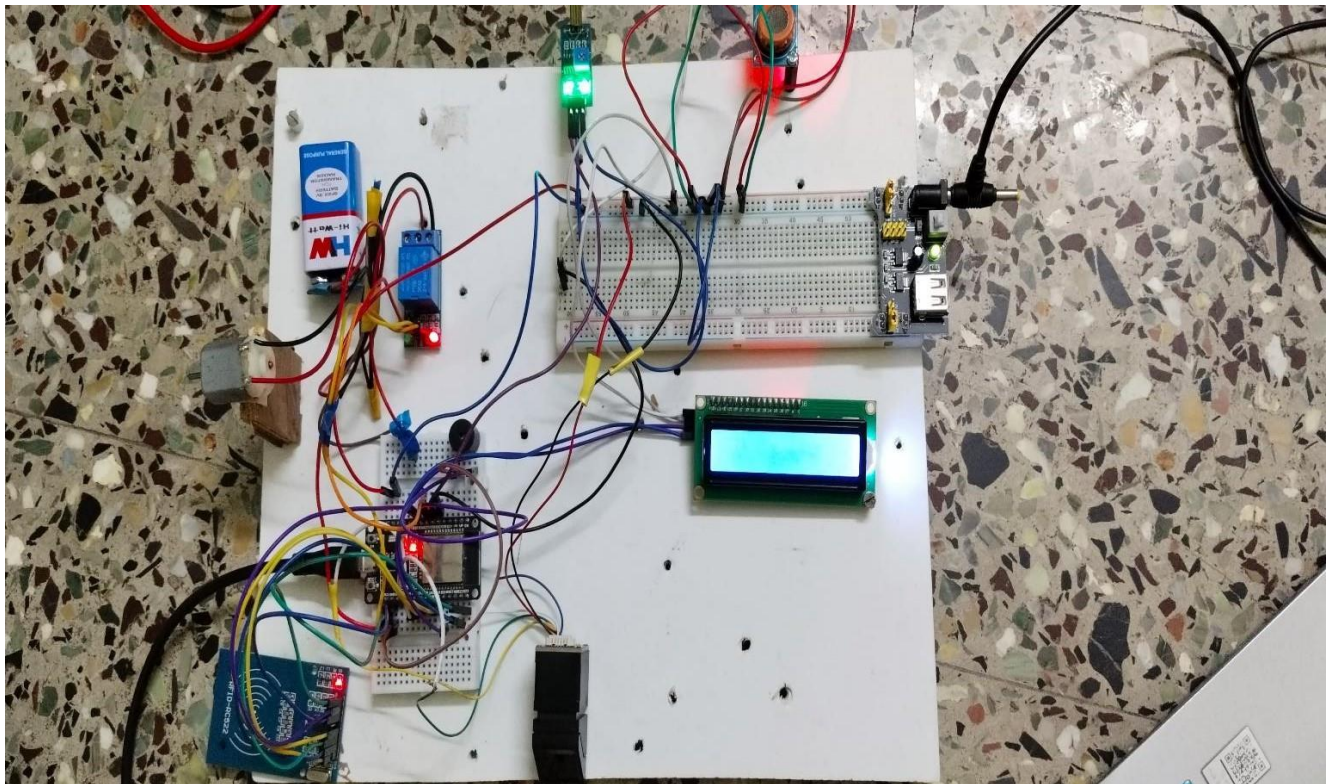


Figure 8.2.2: DC Motor Ignited

CHAPTER 9

APPLICATIONS, ADVANTAGES AND LIMITATIONS

9.1 APPLICATIONS

This multi-level vehicle security system has several practical applications:

1. **Personal Vehicles:** Enhancing the security of private cars, motorcycles, and trucks against theft and unauthorized use. Provides peace of mind to owners with remote monitoring.
2. **Commercial Fleet Management:** For companies with a fleet of vehicles (e.g., delivery services, taxi companies), this system can ensure that only authorized drivers operate specific vehicles, track usage (via notifications), and potentially reduce insurance premiums by demonstrating enhanced security. The alcohol detection feature is particularly valuable for commercial driver safety.
3. **Rental Car Agencies:** Can be adapted to ensure that only the registered renter can operate the vehicle and to monitor for misuse or potential accidents. RFID could be linked to rental agreements.
4. **Heavy Machinery and Equipment:** Securing expensive construction or agricultural machinery that is often left on site and vulnerable to theft or unauthorized operation.
5. **Shared Mobility Services:** Could be integrated into car-sharing or bike-sharing platforms to provide robust user authentication before vehicle use.

9.2 ADVANTAGES

1. **Enhanced Security:** Multi-layer authentication (face, fingerprint, RFID) makes it significantly more difficult for unauthorized individuals to bypass the system compared to single-factor methods.
2. **Increased Driver Safety:** Integration of alcohol detection helps prevent drunk driving. Accident detection (tilt sensor) can provide early alerts.
3. **Remote Monitoring and Control:** IoT integration via Blynk allows owners to receive real-time status updates (e.g., engine start) and remotely disable the engine in emergencies or suspected theft.
4. **User Identification:** Face and fingerprint recognition provide strong, unique identification of the authorized user.
5. **Customizable and Scalable:** The modular design allows for potential future additions or modifications of security layers or sensors.
6. **Deterrent to Theft:** The visible presence of advanced security measures (like an LCD prompt for biometrics) can act as a deterrent.
7. **Reduced Risk of Key-Related Issues:** Mitigates risks associated with lost, stolen, or duplicated physical keys.

9.3 LIMITATIONS

1. Face Recognition Challenges:

- Performance can be affected by extreme lighting conditions (very dark or direct sunlight), significant changes in user appearance (e.g., masks, heavy makeup, different hairstyles not in training), or if the camera is obscured.
- Potential for spoofing with high-quality photos or videos if liveness detection is not implemented (LBPH itself doesn't have strong liveness detection).

2. **Fingerprint Sensor Issues:** Can be unreliable with wet, dirty, or scarred fingers. Quality of the sensor affects performance.

3. **Alcohol Sensor Limitations:** MQ-3 sensors require warm-up time. They can be sensitive to other volatile organic compounds, potentially leading to false positives. Accurate BAC (Blood Alcohol Content) correlation is difficult without precise calibration.

4. **Tilt Sensor Simplicity:** A basic tilt sensor (like SW-520D) might trigger false alarms on steep inclines or very bumpy roads if not calibrated carefully. It cannot distinguish types of impacts.

5. **Power Consumption:** The ESP32 (especially with Wi-Fi active) and various sensors can consume considerable power, which might be a concern for long-term battery operation in a real vehicle if not connected to the vehicle's main power supply with proper management.

CONCLUSION

This project successfully demonstrated the design and implementation of a "Multi-Level Security System for Vehicles" incorporating face recognition, fingerprint scanning, RFID validation, alcohol detection, accident sensing, and IoT-based remote monitoring and control. The system effectively integrates Python for face recognition on a laptop with an ESP32 microcontroller managing subsequent security layers, sensors, and user interaction via an LCD and the Blynk platform.

The layered authentication approach significantly enhances vehicle security compared to traditional systems. The inclusion of safety features like alcohol and accident detection contributes positively to driver and vehicle safety. The IoT capabilities provide valuable real-time awareness and control to the vehicle owner.

While the prototype has certain limitations, primarily related to the practical deployment of laptop-based face recognition in a vehicle and sensor accuracies, it serves as a strong proof-of-concept for a comprehensive and intelligent vehicle security and safety solution. The project successfully met its core objectives by creating a functional system that validates users through multiple checks before granting access and allowing engine operation.

FUTURE SCOPE

1. **Embedded Face Recognition:** Replace the laptop-based face recognition with an embedded solution using a more powerful microcontroller (like ESP32-CAM with limitations, or a Raspberry Pi, Jetson Nano) directly integrated into the vehicle for a standalone system.
2. **Liveness Detection:** Implement liveness detection for the face recognition module to prevent spoofing attacks using photos or videos.
3. **GPS Integration:** Add a GPS module for real-time vehicle tracking, which can be reported via Blynk or SMS in case of theft or accident.
4. **GSM/LTE Connectivity:** Incorporate a GSM/LTE module for sending SMS alerts and enabling remote control even when Wi-Fi is unavailable, providing more robust connectivity.
5. **Advanced Accident Detection:** Use a more sophisticated accelerometer/gyroscope (like MPU6050) and implement algorithms to better differentiate between minor bumps, harsh braking, and actual collisions, reducing false positives from the tilt sensor.
6. **CAN Bus Integration:** Interface with the vehicle's CAN bus (Controller Area Network) to access more vehicle data (speed, RPM, door status) and potentially control more vehicle functions (e.g., door locks, honk horn) directly.
7. **Improved Alcohol Sensor:** Explore more accurate and reliable alcohol sensors, possibly with better calibration methods or fuel-cell based sensors.
8. **Multi-User Management:** Develop a more sophisticated system for managing multiple authorized users with different permission levels, easily configurable by the primary owner.
9. **Power Optimization:** Implement power-saving modes and optimize code for reduced energy consumption, especially if intended for battery-powered standalone operation.
10. **Voice Commands/Feedback:** Integrate voice recognition for commands and voice synthesis for feedback to enhance user interaction.
11. **Encrypted Communication:** Ensure all sensitive data communication, especially over wireless channels, is robustly encrypted.

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Appendix A- Datasheets

1. DATA SHEET of ESP32-WROOM-32

Parameter	Specification
Microcontroller	Tensilica Xtensa LX6 dual-core (240 MHz)
RAM	520 KB SRAM
Flash Memory	4 MB (external)
Operating Voltage	3.0V – 3.3V
I/O Voltage Tolerance	Not 5V tolerant (3.3V logic only)
GPIOs	34 (multiplexed with ADC, DAC, PWM, etc.)
Communication Interfaces	UART, SPI, I2C, I2S, CAN, PWM
Wireless Connectivity	Wi-Fi (802.11 b/g/n), Bluetooth v4.2 BR/EDR + BLE
Operating Temperature	-40°C to +85°C
Power Modes	Deep Sleep, Light Sleep, Modem Sleep
Package	SMD Module
Dimensions	~18 mm × 25.5 mm

2. DATA SHEET of RC522 RFID READER

Parameter	Specification
Operating Frequency	13.56 MHz
Supported Protocols	ISO/IEC 14443 Type A
Communication Interface	SPI (default), I2C, UART
Voltage Supply	2.5V – 3.3V
Current Consumption	~13 mA (active), < 10 μ A (power-down)
Read Range	Up to 10 cm (depends on antenna and card)
Data Transfer Rate	106 kbit/s
Antenna	Integrated (PCB trace)

Operating Temperature	-25°C to +85°C
Dimensions	~40 mm × 60 mm
Datasheet	MFRC522 Datasheet (PDF)

3. DATA SHEET of R307/R503 FINGERPRINT SENSOR

Parameter	Specification
Sensor Type	Optical
Image Size	256 × 288 pixels
Template Storage	Up to 1000 templates (R307) / up to 3000 (R503)
Matching Speed	< 1 sec
FAR / FRR	FAR < 0.001%, FRR < 0.1%
Operating Voltage	3.6V – 6V
Interface	UART TTL
Communication Speed	Default: 57600 bps
Scanning Window	~18 mm × 22 mm
Built-in Flash	Yes
Dimensions (R307)	~44.6 × 20.4 × 21.5 mm
Datasheets	R307 Datasheet (PDF) R503 Info (PDF)

4. DATA SHEET OF MQ-3 ALCOHOL SENSOR

Parameter	Specification
Detection Range	0.05 mg/L – 10 mg/L Alcohol in air
Heater Voltage	5.0V ± 0.2V
Load Resistance	200 kΩ – 10 kΩ
Sensitivity	High sensitivity to ethanol, benzene
Response Time	< 10 seconds
Recovery Time	< 30 seconds
Output Signal	Analog (voltage increases with alcohol concentration)
Preheat Time	≥ 20 seconds

Operating Temperature	-10°C to +70°C
Power Consumption	≤ 750 mW
Datasheet	MQ-3 Datasheet (PDF)

5. DATA SHEET OF 1602 I2C LCD

Parameter	Specification
Display Format	16 characters $\times$ 2 lines
Interface	I2C (via PCF8574 I/O expander)
Input Voltage	5V
Logic Voltage	5V (but works with 3.3V MCU using level shifter or voltage divider)
Backlight	LED backlight (usually white or green)
Address (default)	0x27 or 0x3F (can be changed via solder pads)
Contrast Control	Via onboard potentiometer
Module Dimensions	~ 80 mm $\times$ 36 mm
Operating Temperature	0°C to 50°C
Datasheet	1602 LCD Reference

6. Data sheet of SW-520D Tilt Sensor

Parameter	Specification
Type	Ball-based tilt/vibration sensor
Operating Voltage	3.3V – 5V
Output Type	Digital (HIGH when flat, LOW when tilted)
Response Angle	~15° – 45° tilt
Debouncing	May require software debouncing
Operating Temperature	-20°C to +70°C
Dimensions	~15 mm length
Datasheet	SW-520D Datasheet

7. Data sheet of Relay Module

Parameter	Specification
Relay Type	Electromechanical
Operating Voltage	5V DC
Trigger Level	Low level trigger (active LOW)
Max Switching Voltage	AC: 250V @10A / DC: 30V @10A
Optocoupler Isolation	Yes
Input Pin	IN (trigger), VCC, GND
Load Type	AC or DC devices
Module Size	~50 mm × 25 mm
LED Indicator	Yes (shows relay ON/OFF status)
Datasheet	Relay Module Reference

Appendix B - Code

Dataset collection.py

```
import cv2

import random

cam=cv2.VideoCapture(0)

cascade=cv2.CascadeClassifier("haarcascade_frontalface_default.xml")

while True:

    flag,frame=cam.read()

    test_image_gray=cv2.cvtColor(frame,cv2.COLOR_BGR2GRAY)

    faces=cascade.detectMultiScale(test_image_gray,1.1,5)

    print(faces)

    if len(faces) >0 :

        x,y,w,h=faces[0]

        cv2.rectangle(frame,(x,y),(x+w,y+h),(255,0,0),2)

        cv2.imshow("testimage",frame)

        k=cv2.waitKey(2)

        if k==ord('q'):

            break

        if k==ord('s'):

            roi=frame[y:y+h,x:x+w] # face cropping

            roi=cv2.resize(roi,(300,300))

            n=random.randint(1,200)

            filename=f"./dataset/0/person{n}.jpg"
```

```
cv2.imwrite(filename,roi)
```

```
cam.release()
```

Training.py

```
"""
in this file we extract features of
dataset and train the model using LBPH

"""
from os import listdir
import cv2
import numpy as np
recog=cv2.face.LBPHFaceRecognizer_create()
root_dir="./dataset"
features=[]
label=[]

i=0
for subfolder in listdir(root_dir):
    folder_path=f"{root_dir}/{subfolder}"
    print(f"-----{folder_path}----- ")
    for file in listdir(folder_path):
        file_path=f"{folder_path}/{file}"
        print(file_path)
        image=cv2.imread(file_path,0)

        features.append(image)
        label.append(i)
        # print(image)
        # cv2.imshow("img",image)
        # cv2.waitKey()

    i+=1
```

Prediction.py

```
import cv2
import numpy as np
from apidemo import insert

recog=cv2.face.LBPHFaceRecognizer_create()
cam=cv2.VideoCapture(0)

names={
    0:"K.B.raj",
    1:"Suchitra",
    2:"Shwetha"
}

recog.read("facemodel.yml")
cascade=cv2.CascadeClassifier("haarcascade_frontalface_default.xml")

#above line reads the model

while True:

    # test_image=cv2.imread("./dataset/0/person2.jpg")

    flag,test_image=cam.read()

    if flag:

        gray=cv2.cvtColor(test_image,cv2.COLOR_BGR2GRAY)

        faces=cascade.detectMultiScale(gray,1.1,5)
```

```
if len(faces) > 0:
```

```
    x,y,w,h=faces[0]
```

```
    cv2.rectangle(test_image,(x,y),(x+w,y+h),(255,0,0),2)
```

```
    roi=gray[y:y+h,x:x+w]
```

```
    roi=cv2.resize(roi,(300,300))
```

```
    id,confi=recog.predict(roi)
```

```
    result=f"id : {id} , confi : {confi}"
```

```
    msg="Welcome"
```

```
    if confi < 30:
```

```
        detected_name=names[id]
```

```
        flag=insert(detected_name)
```

```
        if flag=="1":
```

```
            msg="Attendance for the today already recorded"
```

```
        else:
```

```
            msg="Attendance taken,Thank you"
```

```
    else:
```

```
        detected_name="Un known"
```

```
    text = detected_name
```

```
    font = cv2.FONT_HERSHEY_SIMPLEX
```

```
    position = (50, 50) # (x, y) coordinates
```

```
    font_scale = 1
```

```
    font_color = (0, 0, 255) # Red color in BGR
```



```
thickness = 2
line_type = cv2.LINE_AA

# Put text on the image
cv2.putText(test_image, text, position, font, font_scale, font_color, thickness, line_type)

cv2.putText(test_image, msg, (50,100), font, font_scale,(255,0,0), thickness, line_type)

print(result)

cv2.imshow("Result of prediction",test_image)
k=cv2.waitKey(2)

if k==ord('q'):
    break

cam.release()
```

COURSE OUTCOMES

On completion of this course, students will be able to

- Get exposed to literature available for research challenges in interdisciplinary fields.
- Get an insight of real-world problems, challenges and related engineering issues to broader societal context.
- Implement their idea using modern tools & new technologies.
- Demonstrate professionalism with ethics, individual & team management and effective communication skills to comprehend & to write reports.

COST ESTIMATION

BUDGET	AMOUNT
Materials / Consumables	8,500
b) Labor	3,000
c) Travel	2,000
d) Miscellaneous	2,000
Total	15,500

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